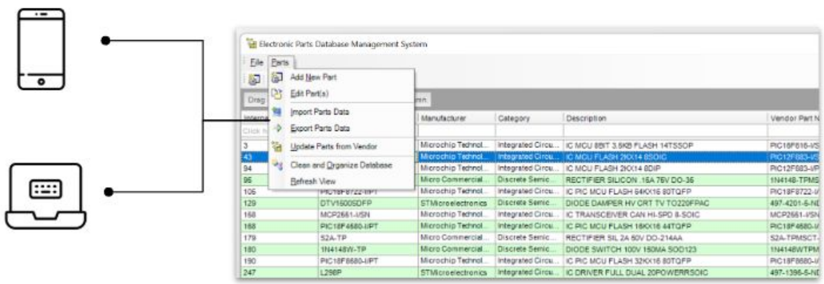
Team 45: LES Parts Database UI
Bi-Weekly Update 4

Anthony Beneventi, Jamaal Bloom, Jaswin Jabbal
Sponsor: John Lusher
TA: Sabyasachi Gupta

Project Summary

- The current database requires manually updating parts and limits multiple users working at the same time.
- Develop a web and mobile application that will allow for the auto population and filtering of part information



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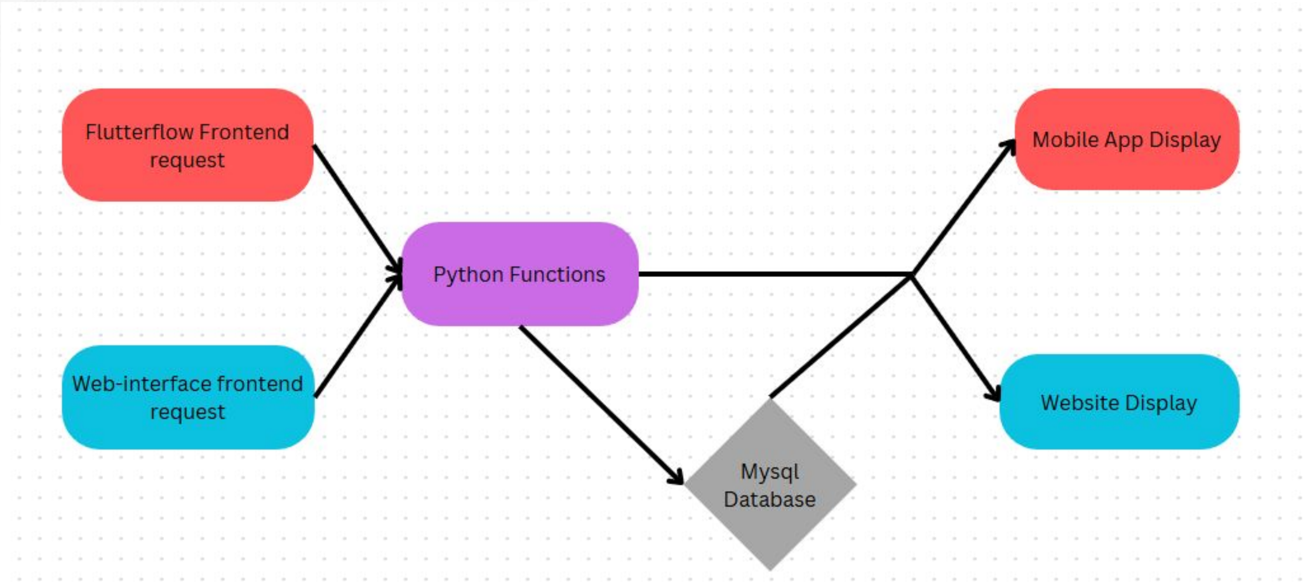
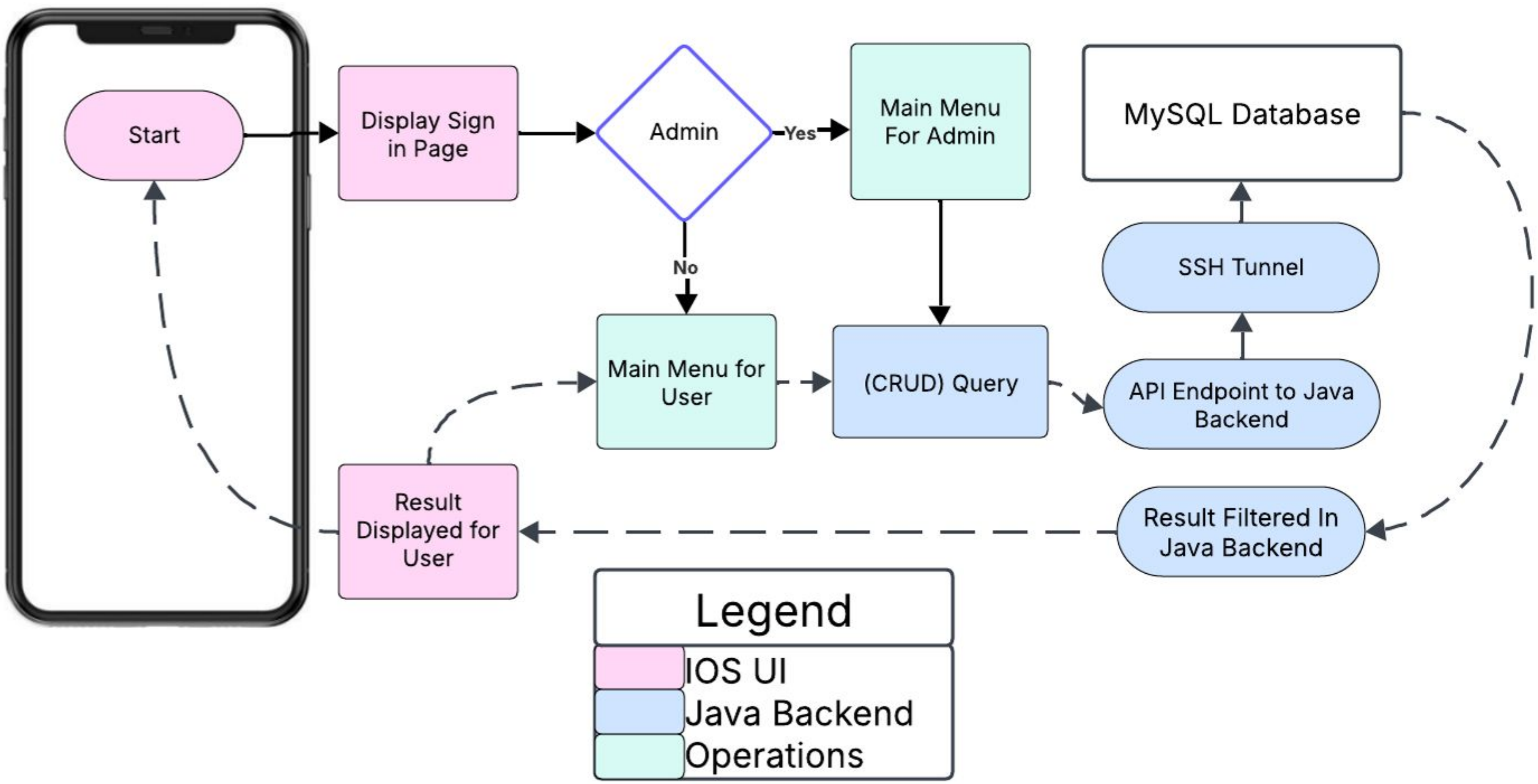
Project/Subsystem Overview



Team:
Jamaal:
Anthony:
Jaswin:

Integrated System Diagram



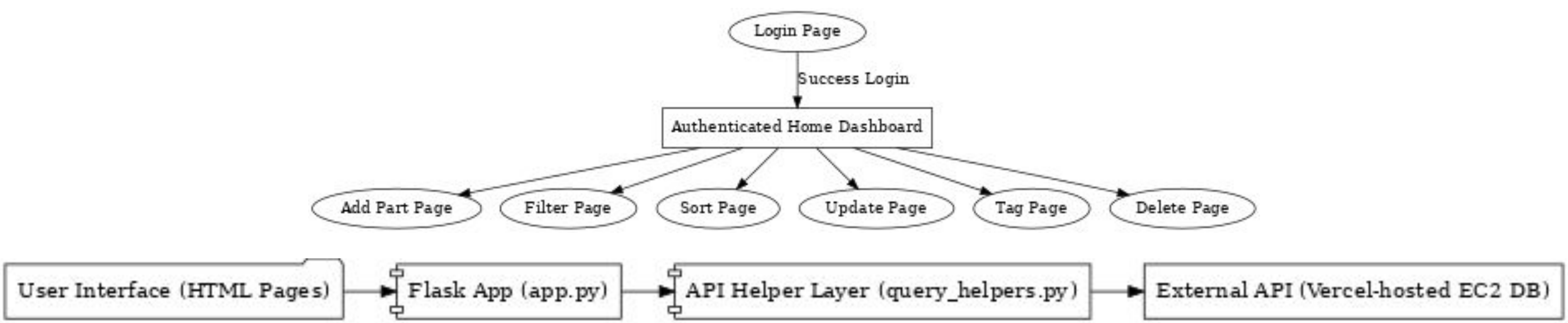
Project Timeline

Subsystem Designs and Testing (completed 1/27)	Start work on a singular database (completed 2/20)	Connect backend API to frontend (to complete by 2/27)	General Integration (to complete by 3/6)	System Test (to complete by 3/17)	Validation (to complete by 3/31)	Demo and Report (to complete by 4/21)

Web Application UI

Jaswin Jabbal

Accomplishments since Update 3 30+ hrs of effort	Ongoing progress/problems and plans until the next presentation
- New UI with working add, filter, sort, update, search functions **Completed testing/validation of Jamaal's backend query functions - next I will be replacing the coded functions with API calls, completing integration.**	- API integration validation **I am currently working on the transition from Jamaal's query functions to his API** - Deployment Environment by next update **Gunicorn, Docker build+run, etc.**



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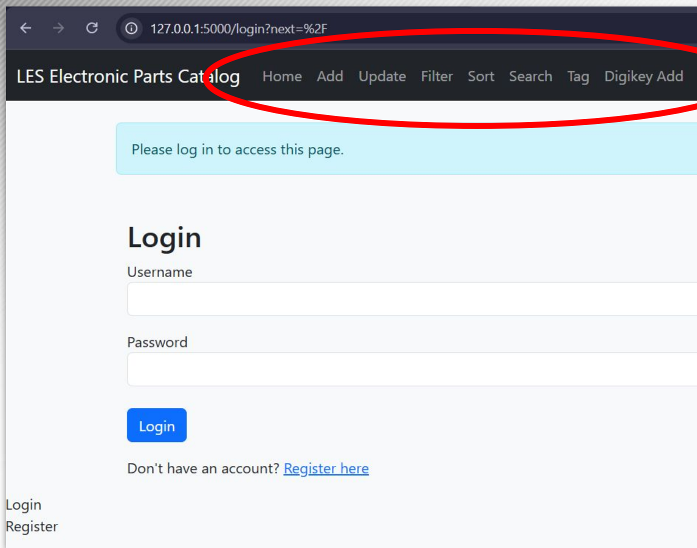
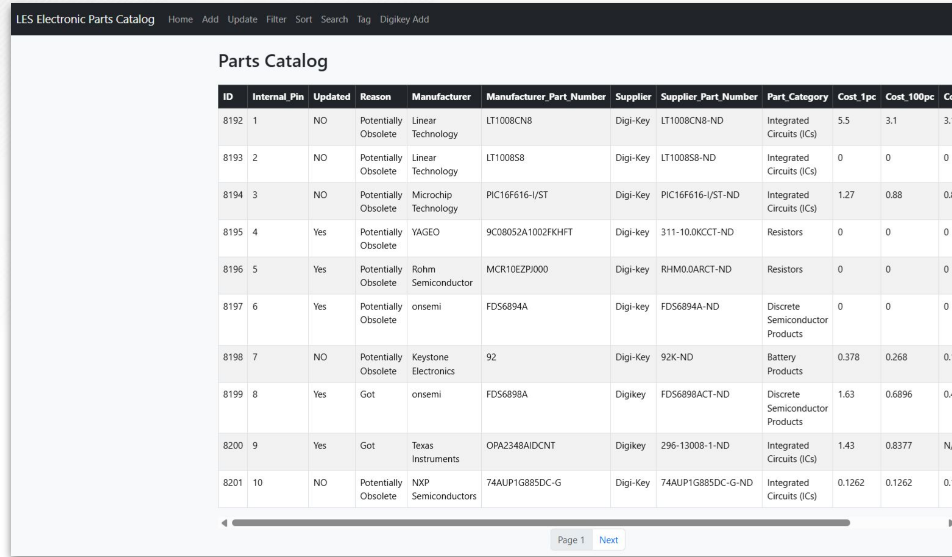


Working CRUD functions



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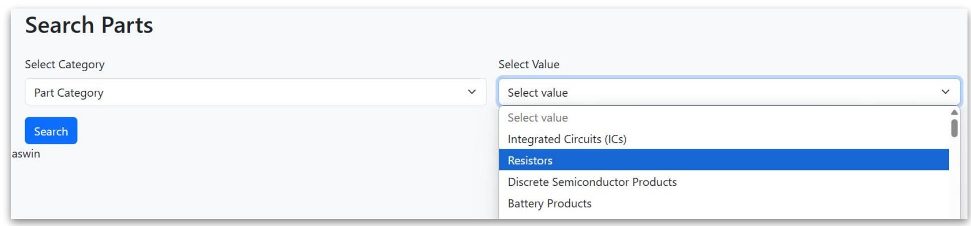
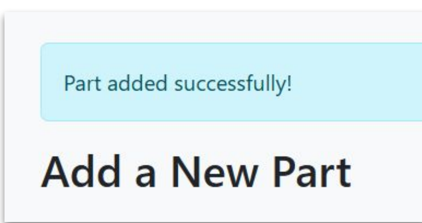
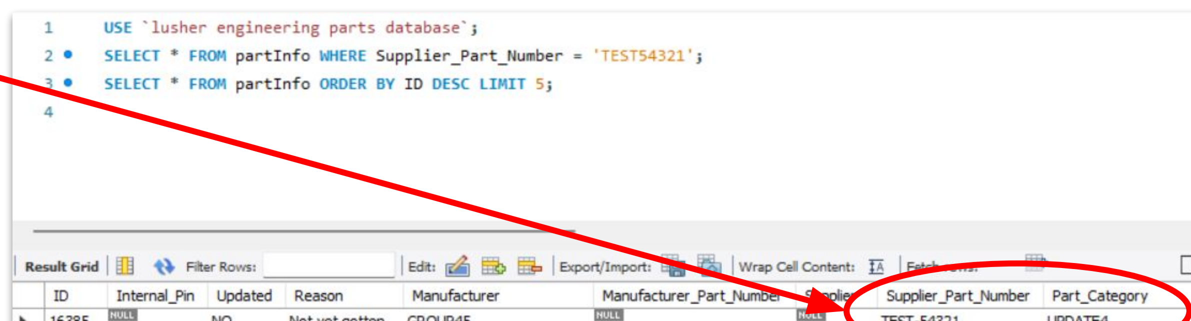
8192	1	NO	Potentially Obsolete	Linear Technology	LT1008CN8	Digi-Key	LT1008CN8-ND	Integrated Circuits (ICs)	5.5	3.1	3.1
8193	2	NO	Potentially Obsolete	Linear Technology	LT1008S8	Digi-Key	LT1008S8-ND	Integrated Circuits (ICs)	0	0	0
8194	3	NO	Potentially Obsolete	Microchip Technology	PIC16F616-I/ST	Digi-Key	PIC16F616-I/ST-ND	Integrated Circuits (ICs)	1.27	0.88	0.8
8195	4	Yes	Potentially Obsolete	YAGEO	9C08052A1002FKHFT	Digi-key	311-10.0KCCT-ND	Resistors	0	0	0
8196	5	Yes	Potentially Obsolete	Rohm Semiconductor	MCR10EZPJ000	Digi-key	RHM0.0ARCT-ND	Resistors	0	0	0
8197	6	Yes	Potentially Obsolete	onsemi	FDS6894A	Digi-key	FDS6894A-ND	Discrete Semiconductor Products	0	0	0
8198	7	NO	Potentially Obsolete	Keystone Electronics	92	Digi-Key	92K-ND	Battery Products	0.378	0.268	0.1
8199	8	Yes	Got	onsemi	FDS6898A	Digikey	FDS6898ACT-ND	Discrete Semiconductor Products	1.63	0.6896	0.4
8200	9	Yes	Got	Texas Instruments	OPA2348AIDCNT	Digikey	296-13008-1-ND	Integrated Circuits (ICs)	1.43	0.8377	N/
8201	10	NO	Potentially Obsolete	NXP Semiconductors	74AUP1G885DC-G	Digi-Key	74AUP1G885DC-G-ND	Integrated Circuits (ICs)	0.1262	0.1262	0.1

New landing page includes pagination for current DB view



Dynamic search for parts





Python Backend

Jamaal Bloom

Accomplishments since Update 3 50 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Switched to using a EC2 Database • Updated code to completely work with the EC2 connection • Tested functions on partners' frontend	• Potentially switch back to SSH • Work alongside teammates to have consistent formatting of outputs • Return correct error messages when errors occur.

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Python Backend

```python
def Add():
    url = 'https://404-app-integration.vercel.app/Add'
    headers = {"Authorization" : "Bearer iWtZuwF0U3wwxs4U4Jfm8Z6b"}
    data = {'partnumber': '1N4448XTPMSCT-ND'}          #Partnumber to be added, Currently needs to be supplier part number and specifacl
    response = requests.post(url, data=data, headers=headers)
    return(response.text)

def Tag():
    url = 'https://404-app-integration.vercel.app/Tag'
    headers = {"Authorization" : "Bearer iWtZuwF0U3wwxs4U4Jfm8Z6b"}
    data = {'partnumber': '1N4448XTPMSCT-ND',       #Partnumber where tag is to be added
            'tag' : "Test 4"}                      #Tag to be added
    response = requests.post(url, data=data, headers=headers)
    return(response.text)

def Update():
    url = 'https://404-app-integration.vercel.app/Update'
    headers = {"Authorization" : "Bearer iWtZuwF0U3wwxs4U4Jfm8Z6b"}
    data = {'lower' : '0',                         #lower bound
            'upper' : '10'}                        #upper bound, Recommended don't do more then like 10 as it may time out
    response = requests.post(url, data=data, headers=headers)
    return(response.text)
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ID	Internal_Pin	Updated	Reason	Manufacturer	Manufacturer_Part_Number	Supplier	Supplier_Part_Number	Part_Category	Cos
1	1	NO	Potentially Obsolete	Linear Technology	LT1008CN8	Digi-Key	LT1008CN8-ND	Integrated Circuits (ICs)	5.5
2	2	NO	Potentially Obsolete	Linear Technology	LT1008S8	Digi-Key	LT1008S8-ND	Integrated Circuits (ICs)	0
3	3	NO	Potentially Obsolete	Microchip Technology	PIC16F616-I/ST	Digi-Key	PIC16F616-I/ST-ND	Integrated Circuits (ICs)	1.27
4	4	Yes	Potentially Obsolete	YAGEO	9C08052A1002FKHFT	Digi-key	311-10.0KCCT-ND	Resistors	0
5	5	Yes	Potentially Obsolete	Rohm Semiconductor	MCR10EZPJ000	Digi-key	RHM0.0ARCT-ND	Resistors	0
6	6	NO	Potentially Obsolete	Fairchild Semiconductor	FDS6894A	Digi-Key	FDS6894A-ND	Discrete Semiconductor Products	0
7	7	NO	Potentially Obsolete	Keystone Electronics	92	Digi-Key	92K-ND	Battery Products	0.37
8	8	NO	Potentially Obsolete	Fairchild Semiconductor	FDS6898A	Digi-Key	FDS6898ACT-ND	Discrete Semiconductor Products	1.28
9	9	Yes	Got	Texas Instruments	OPA2348AIDCNT	Digikey	296-13008-1-ND	Integrated Circuits (ICs)	1.43

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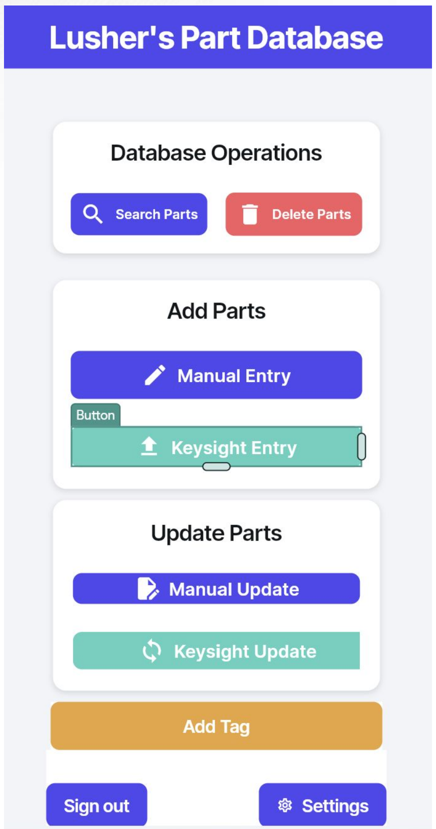
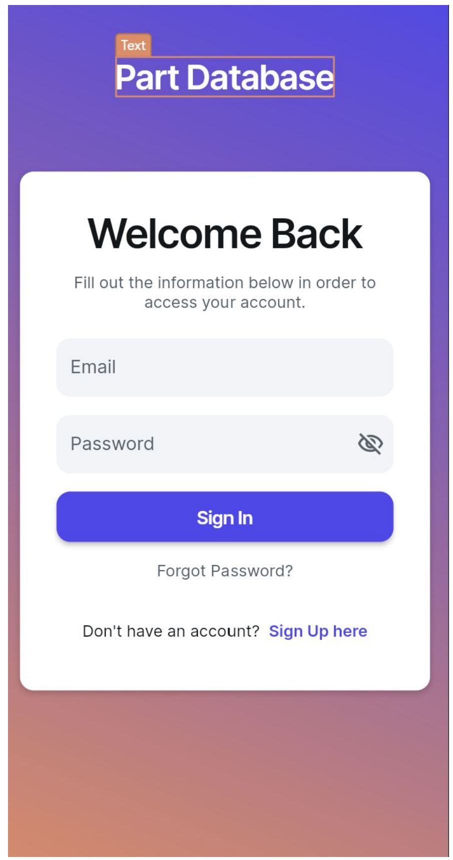
IOS Application

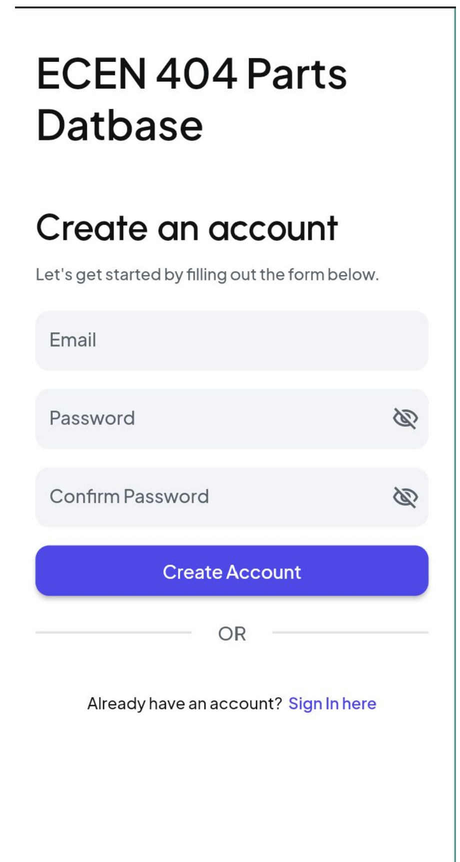
Anthony Beneventi

Accomplishments since Update 3 45+ hrs of effort	Ongoing progress/problems and plans until the next presentation
• Integrated with SSH Tunnel & Dr. Lusher’s Database • Added Jamaal’s API Function & Tag Functions to App UI. • Added User Authority • Deployed Online	• Validation Tests • Verify Multiple User Access • Improve UI & Code Appearance • Final Improvements • Deployed to App Store

IOS Subsystem







Execution Plan

Task Assignments	Owner	1/13/25	1/20/25	1/27/25	2/3/25	2/10/25	2/17/25	2/24/25	3/3/25	3/10/25	3/17/25	3/24/25	3/31/25	4/7/25	4/14/25	4/21/25	4/28/25	Due Date
Finish Subsystems	Team																	
Status Update #1	Team																	1/27/25
Convert code to callable API	Jamaal																	
New models based on DB requirement	Jaswin																	
Fix Update/Insert	Anthony																	
Status Update #2	Team																	2/10/25
Connect with online database	Jamaal																	
Routes based on new models	Jaswin																	
Integrate Database	Anthony																	
Status Update #3	Team																	2/24/25
Format frontend and backend requests	Jamaal																	
Data Consistency	Jaswin																	
Consolidate App	Anthony																	
Finish general integration	Team																	
Status Update #4	Team																	3/17/25
Implement failsafes	Jamaal																	
Deployment	Jaswin																	
System validation	Anthony																	
Staus Update #5	Team																	3/31/25
Finalize code	Jamaal																	
Deployment	Jaswin																	
Optimize code	Anthony																	
Final Presentation	Team																	4/14/25
Showcase/ Demo	Team																	4/24/25

Key	
Planned	
In Progress	
Complete	
Behind Schedule	

Validation Plan

Pararagph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.4	Database Sync	The web and mobile applications uses the same database and changes visible from both	Attempt changes from multiple devices and ensure consistency	Untested	Jamaal
3.2.1.1	API Correctness	The API pulls correct information from the supplier	Compare recieved information with suppliers website	TESTED	Jamaal
3.2.6.2	Request Correctness	The backend correctly recieves and handles the frontend requests	Send frontend request from both applications and examine results	Untested	Jamaal
3.2.1.3	Multiple Requests	Application can handle multiple requests at the same time	Send multiple requests	Untested	Jamaal
3.2.2.1	Model Validation	All DB models align correctly with the updated schema used by Jamaal	Compare Flask SQLAlchemy models against the latest database schema	TESTED	Jaswin
3.2.4.1	Route Functionality	Form submission routes to the correct display without errors	Manually fill out and submir forms in the application, verifying data correctness	TESTED	Jaswin
3.2.4.2	Frontend Data Consistency	Submitted data is correctly displayed on a corresponding UI template	Manual validation by cross-checking displayed records with DB entries	TESTED	Jaswin
3.2.6.1	Deployment Environment	All required dependencies are correctly installed and configured on the end machine	Validate with Docker, verify installation logs, ensure WGSI server	Untested	
3.2.1.2	CRUD Operation Test	Pulling apart information from the MySQL Database	Access a local MySQL Server with dummy datta	TESTED	Anthony
3.2.1.4	Updated Database	The API Correctly updates the entire database	Checking previous and updated dababases to ensure completion	TESTED	Anthony
3.2.1.5	Request Correctness	Having all requested components, such as User authority	Giving specific permission to individual User admins	Untested	Anthony
3.2.1.3	Application Deployed	Uploaded on the cloud and ready for use	Able to be sent for beta testing	Untested	Anthony

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Questions?